

PSP - Outbox On-boarding

Create a table for the psp_outbox under pspadm schema .

```
--Connect pspadm_owner and create tables

CREATE TABLE IF NOT EXISTS pspadm.psp_outbox
(
    event_id                uuid                NOT NULL,
    topic                   VARCHAR(255)        NOT NULL,
    partition_key            VARCHAR(255)        NOT NULL,
    version                 INTEGER              NOT NULL,
    utc_time                 TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT (CURRENT_TIMESTAMP AT TIME ZONE
'UTC'),
    intuit_entity_type       VARCHAR(255)        NOT NULL,
    event_recorded_timestamp VARCHAR(32)          NOT NULL,
    region                  VARCHAR(255)        NOT NULL,
    headers                 JSONB,
    payload                 JSONB,
    outbox_seq              CHARACTER VARYING(255) NOT NULL,
    creator_id              CHARACTER VARYING(30),
    created_date            TIMESTAMP(6) WITHOUT TIME ZONE NOT NULL,
    modifier_id             CHARACTER VARYING(30),
    modified_date           TIMESTAMP(6) WITHOUT TIME ZONE NOT NULL,
    realm_id                SMALLINT NOT NULL DEFAULT - 1,
    PRIMARY KEY (event_id, utc_time)
) PARTITION BY range(utc_time);

CREATE INDEX IX_intuit_entity_type ON pspadm.psp_outbox(intuit_entity_type);
CREATE INDEX IX_event_recorded_timestamp ON pspadm.psp_outbox(event_recorded_timestamp);

CREATE TABLE IF NOT EXISTS pspadm.outbox_heartbeat
(
    utc_time                TIMESTAMP WITHOUT TIME ZONE NOT NULL DEFAULT (CURRENT_TIMESTAMP AT TIME ZONE 'UTC') PRIMARY
KEY,
    is_realtime             BOOLEAN                  NOT NULL
);
```

1. Add pg_cron to DB cluster parameter group (parameter: shared_preload_libraries--> add pg_cron. and rds.logical_replication to 1
2. DB restart required.

```

#partman extensions setup connect ppspg02 DB

create schema partman;
CREATE EXTENSION pg_partman WITH SCHEMA partman;

#connect postgres with postgres DB and setup cron.
postgres=> CREATE EXTENSION pg_cron;
postgres=>SELECT cron.schedule('outbox_partitioned_job', '20 22 * * *', $$SELECT partman.run_maintenance
('pspadm.psp_outbox')$$);
postgres=>UPDATE cron.job SET database = 'ppdspg02' WHERE jobid = 34;--> SELECT * FROM cron.job;

--login postgres and Create user
CREATE ROLE data_capture_role WITH PASSWORD 'XXXX' LOGIN;
GRANT rds_replication TO data_capture_role;
GRANT CONNECT ON DATABASE ppspg02 TO data_capture_role;
GRANT USAGE ON SCHEMA pspadm TO data_capture_role;
GRANT SELECT ON pspadm.psp_outbox TO data_capture_role;
GRANT SELECT, INSERT, DELETE ON pspadm.outbox_heartbeat TO data_capture_role;

--Create publication
CREATE PUBLICATION data_capture_outbox_publication_e2e
    WITH (publish = 'insert');

SELECT partman.create_parent(
    p_parent_table => 'pspadm.psp_outbox',
    p_control => 'utc_time',
    p_type => 'native',
    p_publications => ARRAY['data_capture_outbox_publication_e2e'],
    p_interval=> 'daily',
    p_start_partition=> (CURRENT_TIMESTAMP AT TIME ZONE 'UTC')::text,
    p_premake => 2);

UPDATE partman.part_config
    SET infinite_time_partitions = true,
        retention = '7 days',
        retention_keep_table=false
    WHERE parent_table = 'pspadm.psp_outbox';

ALTER PUBLICATION data_capture_outbox_publication_e2e
ADD TABLE pspadm.outbox_heartbeat;

###select * from pg_publication_tables where pubname='data_capture_outbox_publication_e2e';###SELECT * FROM
cron.job; ###SELECT * FROM cron.job_run_details;

```

Outbox setup: <https://github.intuit.com/pages/ceres/documentation/docs/clients/managed-outbox/onboard-for-postgres/>

Outbox table rename:<https://github.intuit.com/pages/ceres/documentation/docs/clients/managed-outbox/onboard-for-postgres/outbox-table-schema-migration>